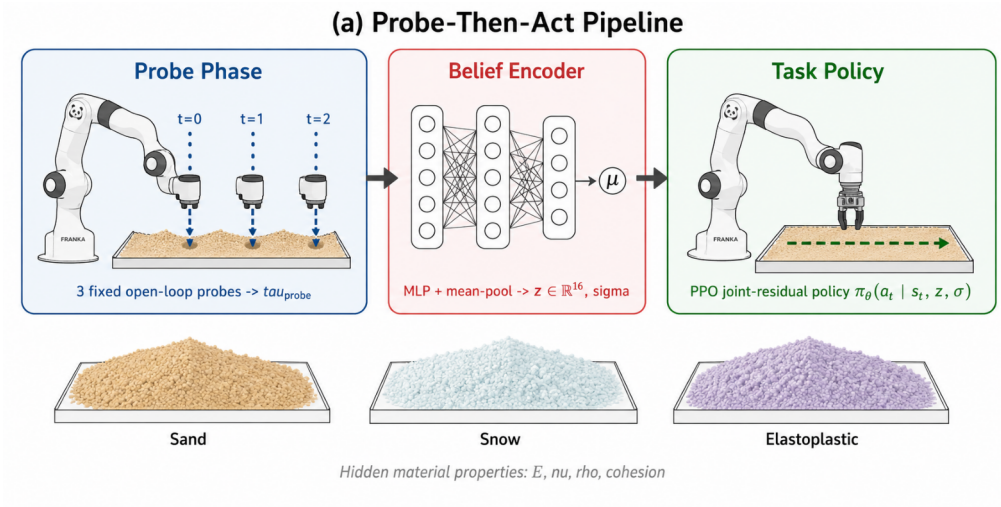
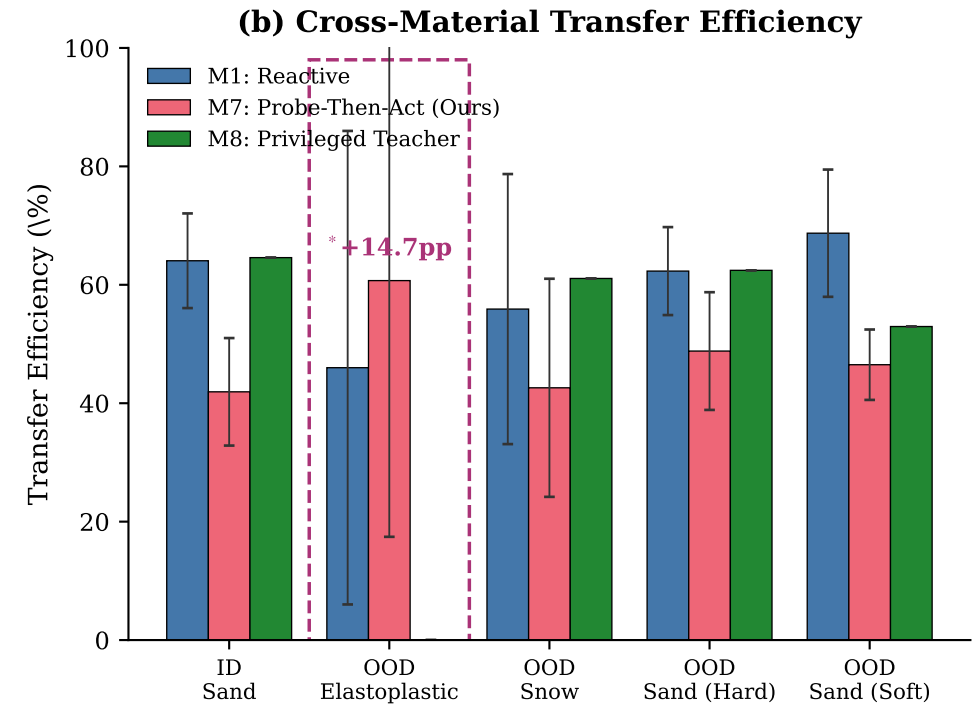

(a) Probe-Then-Act Pipeline
Probe Phase
t=0
t=1
t=2
3 fixed open-loop probes -> tau_{probe}
Belief Encoder
μ
MLP + mean-pool -> $z \in \mathbb{R}^{16}$, sigma
Task Policy
PPO joint-residual policy $\pi_\theta(a_t \mid s_t, z, \sigma)$
Sand
Snow
Elastoplastic
Hidden material properties: E, nu, rho, cohesion


(b) Cross-Material Transfer Efficiency
M1: Reactive
M7: Probe-Then-Act (Ours)
M8: Privileged Teacher
Transfer Efficiency (%)
0
20
40
60
80
100
*+14.7pp
ID Sand
OOD Elastoplastic
OOD Snow
OOD Sand (Hard)
OOD Sand (Soft)